

# Cover Letter

Dear Professor,

I am S. H. M. Muntasir Rahi, currently pursuing an M.S. in Mechanical Engineering at Clarkson University with a graduate GPA of 3.945/4.00. I am writing to express my interest in Ph.D. research opportunities in mechanical engineering, materials science, advanced manufacturing, and data-driven material characterization.

My research interests lie in ultrasonic non-destructive evaluation, signal processing, physics-informed machine learning, additive manufacturing, and multi-scale material characterization. I am especially interested in understanding structure–property relationships, microstructural evolution, material degradation, and damage mechanisms in complex engineering materials.

My current research focuses on ultrasonic non-destructive characterization and physics-informed machine learning for material property evaluation. I work on extracting microstructural, mechanical, and micro-viscoelastic information from ultrasonic wave propagation, attenuation, dispersion, and spectral response features. My projects involve experimental design, waveform acquisition, feature extraction, MATLAB/Python-based signal processing, data analysis, and machine learning model development.

I am also working on ultrasonic characterization of pharmaceutical granular composites under moisture and thermal exposure, where I study irreversible microstructural and micro-viscoelastic changes using time-domain, frequency-domain, attenuation, and dispersion-based analysis. In addition, I have experience with ultrasonic and machine learning-based characterization of 3D-printed PLA structures, focusing on the prediction of manufacturing parameters such as layer height and infill density.

My previous research includes ultrasonic defect detection in metallic materials, which resulted in a conference paper at the ASME International Mechanical Engineering Congress and Exposition. Through these experiences, I have developed a strong foundation in experimental material characterization, signal-based diagnostics, additive manufacturing, and data-driven modeling.

I am motivated to pursue Ph.D. research that combines experiments, computation, and machine learning to study advanced materials, manufacturing processes, and material behavior under complex conditions. My long-term goal is to develop reliable experimental and data-driven frameworks for understanding material performance and supporting the design of resilient materials for engineering applications.

I would greatly appreciate the opportunity to discuss how my background and research interests may align with your group's work. Thank you very much for your time and consideration.

Sincerely,  
S. H. M. Muntasir Rahi